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# Statistical And Machine Learning Analysis Of Customer Responses To Promotions And Discounts

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*Abstract*—The issue of how to attract customers no longer appears to be a primary issue for companies within the digital marketplace. Today's e-commerce companies face an additional challenge – that of optimizing their marketing strategies, so as to minimize the amount of waste incurred in their promotional efforts. This paper presents a solution to the above problem by presenting an analytical framework model, which consists of a combination of both inferential statistics and the latest machine learning techniques to help decode what is influencing customer loyalty and customer churn. This paper utilizes Lasso Regression analysis to identify the parameters that are actually affecting total spend, while using the DP-means cluster analysis technique to perform dynamic audience segmentation, in order to provide a clearer understanding of the way in which different audience segments respond to various promotions. Additionally, this report will detail a technical benchmarking study comparing two leading gradient boosting algorithms (LightGBM and CatBoost) in an effort to determine which is the superior predictive model for determining customer attrition. Based on the experimental findings, the hierarchy of influence between three types of behavioral signal (i.e., tenure, frequency of purchase, and history of recent transactions) has greater predictive capabilities than basic demographic information (i.e., gender and age). Another significant finding is that a "Heavy-Tailed" revenue distribution exists within the data (meaning there are a few individuals who create most of the profit for the business). This finding shifts the emphasis from a quantity-based marketing strategy to a value-based marketing strategy. Therefore, the results provided by this research serve as both a technical & strategic guide for e-commerce stakeholders who would like to shift their focus from using blanket-based discounting to utilizing highly tailored, behaviorally-based engagement.

Keywords—Customer Lifetime Value (CLV), E-commerce Analytics, Predictive Modeling, Coupon Addiction, Customer Attrition, Machine Learning, RFM Segmentation.

## I. Introduction

The intensive digital transformation of the retail industry has 10 led to a striking shift from traditional 10 mass marketing to personalized customer 10 strategies relying on data. In a very competitive online shopping market, businesses are using more and more aggressive marketing tactics, such as flash deals and personalised coupons, to boost sales right away'(Tran Anh et al.[1], 2022). However, the excessive reliance on these incentives without a full awareness can result in a considerable strategic problem: Are these promotional campaigns constructing true loyalty, or are they just attracting deal-seekers who will churn once the promotion is no longer available?

Recent studies and reports have shown that only a marginal 5% increase in customer retention can help boost the profit by 25% to 95% (Reichheld and Schefter [2], 2000). As an obvious result, the main focus of modern business intelligence has moved from the quantity of transactions to the quality of the customer lifecycle, which is represented by CLV (Customer Lifetime Value).

In spite of the abundance of descriptive statistics and static data, many businesses are still struggling to differentiate between sustainable growth and temporary sales spikes caused by coupon availability. This is due to the fact that most research conducted by enterprises is dependent on historical customer behaviors instead of predicting future churn risks through inferential methods and machine learning models. We fill this gap by integrating descriptive statistics & RFM (Recency, Frequency, Monetary) framework, inferential statistics, segmentation algorithm (DP-Means), Lasso Regression, and predictive models (LightGBM and CatBoost) to provide a holistic view of the influence of coupons on customer loyalty.

The primary objective of this study is to analyze the double-edged nature of coupons and to identify how heavy coupon usage may lead to increased churn after promotions end, and to investigate how customer satisfaction shapes the overall impact of coupons on loyalty.

## II. related research

The relationship between price-based promotions and customer loyalty has been widely discussed in many studies. In the Vietnamese e-commerce market, coupons and free shipping are contributing factors to purchase decisions (Tran Anh et al.[1], 2022). Nevertheless, experts warned that a deal-seeking mindset could stem from the over deploying coupons (Garnefeld et al.[3], 2012). This means that when customers become accustomed to discounts, they 7 may switch to 7 competitors offering better deals.

From a technical perspective, the RFM model has been a standard to divide customers into specific groups and detect those who are at risk (Syahra et al.[4], 2025). Thanks to the continuous evolution of advanced technologies, modern machine learning algorithms have been increasingly popular in customer behavior. The key reason for this popularity is their ability to handle large amounts of data and provide better precision than conventional statistical methods.

However, in most prior studies, researchers have only examined coupon usage as a binary variable. In other words, they have tracked whether a customer used a coupon. The main purpose of the study is to analyze the duality of coupons and find out how heavy couponing can lead to higher customer turnover after the end of the promotion period. The role played by happy customers in enhancing the overall impact of coupons on loyalty will be analyzed.

# III. METHODOLOGY

## A. Datasets

To get a full picture of how customers act, we use two datasets from Kaggle. The first dataset is called "Marketing Insights for E-Commerce Company" [5]. It looks at the details of past transactions. Online_Sales (CustomerID, transaction dates, coupon status), Customers_Data (demographics, tenure months), and Discount_Coupon (discount percentages) are some of the most important files. This information is very important for figuring out RFM measures and dividing clients into groups. The second one [6], on the other hand, shows how 5,630 people left the site. CouponUsed (total coupons used), Tenure (how long someone has been a customer), and Churn (the desired variable) are all important factors. Also, Complain and DaySinceLastOrder provide us a caution about loyalty.

These datasets work together to show a direct link between how much you promote something and how long you stay devoted to it. We can also tell the difference between people who are just looking for deals and people who are truly devoted consumers by looking at how often they buy and how often they use discounts.

## B. Data preprocessing

*1) Data cleaning and imputation*

Dataset 1 is clean, with no missing values. Dataset 2, on the other hand, has a minor number of missing values, less than 5.5% at most. Imputation is better than deleting observations because it keeps data from being lost. The sort of feature we choose to use for imputation is what we used. For instance, the median value is used to fill in missing numerical features, and the mode is used to fill in missing categorical features.

*2) Outlier handling and feature engineering*

The IQR (Interquartile Range) approach is used to find outliers by looking for values that are outside the range:
Values between [Q1 - 1.5×IQR and Q3 + 1.5×IQR] are not normal.

Outliers only make up a minor part of all transactions (7.156%), but they bring in a disproportionately huge amount of overall income (33.331%). If you took these values away, the most important financial patterns would be gone. So, these transactions were kept and clearly marked as "VIP" if their value was more than the upper bound figure, and "Regular" if it wasn't.

## C. Descriptive statistics and RFM analysis

*1) RFM modeling and feature engineering*

To quantify the customer value, our research employed the RFM modeling. Although this framework is the standard for measuring behavioral outcomes, it often lacks the ability to reflect the underlying purchase intent in the e-commerce environment with an abundance of promotions. We improved the traditional RFM analysis via specialized feature engineering:

- Recency (R): Simply counting the days since the last transaction. The higher the R number, the more likely it is that people will leave.
- Frequency (F): The number of different invoices. This measure is very important for telling the difference between people who buy things often and those who only buy things when they have coupons.
- Monetary: This is the amount of money left over after taxes and shipping costs. It shows the real value that a consumer brings to the firm.

We handled the data distribution to optimize these features

for the subsequent segmentation algorithm. Both Monetary and Frequency showed significant right-skewness (skewness > 5.5). Therefore, we applied a Log Transformation to these variables so that the variance can be reduced.

*2) Coupon Ratio (CR)*

The fundamental innovation of this research is the introduction of the Coupon_Ratio metric. It is defined as the number of orders that use a promotion code divided by the total number of orders.

CR acts as a representation of the degree of price sensitivity. The CR near 1.0 shows a high level of dependence on coupons, meaning that the relationship of the customer and the business is facilitated by discounts rather than natural loyalty.

*3) Exploratory data analysis (EDA)*

We used visualisation methods to look into the complicated link between applying for a promotion and staying with the company for a long time:

- We can examine the difference between how people buy things when they have a preference and how they buy things when they are looking for a deal by comparing CR with transaction frequency.
- Analyzing churn distribution offers a view that coupon sensitivity is directly proportional to the risk of losing customers. This confirms the hypothesis that loyalty based on short-term discounts is not sustainable.
- When we compare average sales concentration on a variety of product categories, we can identify which categories make customers more active and which ones make them less stable.

*4) Experimental output: Holistic customer profiles*

The combination of EDA and feature engineering leads
to a multidimensional customer profile. Combining RFM scores (how strong the present relationship is), the Coupon_Ratio (how much the promotion depends on the coupon), and Tenure (how long the relationship has lasted) makes a strong base for next phases of predictive modelling.

## D. Inferential statistics

*1) Independent t-test*

The t-test is employed to compare AOV between two groups: customers who utilized coupons and those who did not.

- Null hypothesis ( ): There is no significant difference in AOV between coupon users and non-coupon users.
- Alternative hypothesis ( ): There is a significant difference in AOV between the two groups.

The test used a significant level of α = 0.05. The aim of this test is to determine whether coupons can encourage higher spending.

*2) One-Way ANOVA*

The One-Way ANOVA is an excellent way to see how well different amounts of discounts (10%, 20%, 30%) work when compared to each tier after applying coupons. This will allow us to identify the saturation point where the addition of additional discounts ceases to be of benefit. By analyzing the mean differences between several groups, we aim to verify whether increased discount percentages continue to provide an equivalent economic return or if the revenue growth starts to slow down.

*3) Chi-square test*

The goal of the analysis is to determine if there is a correlation between poor experiences while making a purchase and customer behaviors toward attrition (leaving). This is particularly important as it distinguishes between "true" customers who will stay regardless of the use of coupons and those who are loyal only because they use coupons.

We decided to build a Chi-square test on dataset 2 with the purpose of examining the relationship between categorical variables: *Complain* (service friction) and *Churn* (attrition behavior).

A p-value < 0.05 indicates that customer complaints and the decision to leave are correlated, suggesting that high-quality service is a more effective driver of sustainable loyalty than constant discounting.

*E. Lasso Regression*

In this phase, this research starts to build a predictive baseline model to discover core factors that contribute to customers'values.

*1) Experimental setup and objectives*

This experiment is designed to distinguish strategic behavioral signals from data noise.

- Target variable: A binary variable represents "High-value" customers, defined as those whose transaction frequency (Frequency) exceeds the median. Customers with a frequency above the median were labeled as High-Value (1), while the rest were labeled 0.
- Independent variables: The calculated RFM-C metrics were Recency, Frequency, and Monetary, as well as Coupon_Ratio.

*2) Implementation mechanism*

The implementation mechanism of the experimental workflow follows three main steps:

- Features standardization: Since Lasso is a distance-sensitive mechanism, we needed to standardize the features to a common scale (μ= 0; =1) using StandardScaler so that high magnitude variables (e.g. Monetary) do not put an undue advantage on the distance-based ratio variables (e.g. Coupon_Ratio) in the model.
- L1-regularization: The model was trained with the L1 regularization parameter (α=0.01). The penalty term associated with L1-regularization shrinkage will force all non-essential variables' coefficients toward zero.
- Shrinkage / Feature Selection: The model will shrink features that do not provide useful information to the target variable, as a result of the L1-penalty/regularization, which will prevent overfitting of Lasso due to giving less weight to insignificant predictors.

*F. DP-Means*

We aim to replace subjective confirmation with a segmentation model that uses real data, so DP-Means was utilized. This method lets the data decide how many categories are best depending on how people buy things and how sensitive they are to coupons.

*1) Feature scaling and standardization*

Before performing the DP-Means algorithm, the input metrics must be Z-Score standardised. These metrics are Recency (R), Frequency (F), Monetary (M), and Coupon_Ratio. This is because each of these measures is measured in a different way and needs to be changed to a similar way:

This change puts all the features on the same scale, with a mean of 0 and a standard deviation of 1.

*2) Optimization via the Penalty Parameter (λ)*

Unlike K-Means, which requires pre-determining the number of clusters (K) using the Elbow Method, DP-Means is an effective option that optimizes the Penalty Parameter (λ).

- Logic of λ: This acts as a threshold that decides whether a new cluster will be created. If a customer's behavior is extremely distinct from the current cluster, meaning that their distance from existing centroids exceeds λ, the algorithm will detect and automatically create a new cluster.
- Experimental output: By tuning λ as opposed to forcing the data into pre-set groups, the model was able to discover 8 distinct customer groups. This gives a far higher level of detail, letting you see little but essential changes in behaviour.

*3) Experimental output: A general view of customer segments*

TABLE I. BEHAVIORAL PROFILES OF THE 8 IDENTIFIED STATISTICAL CLUSTERS

| Cluster ID | Profile | Description |
|---|---|---|
| **Cluster 0** | Hibernating | High Recency / Low Frequency; inactive for >250 days. |
| **Cluster 1** | Deel seekers | High CR (>0.65); strictly price-sensitive customers. |
| **Cluster 2** | Stable | Moderate R and M; predictable shopping habits. |
| **Cluster 3** | Elite VIPs | Extreme M / Low R; top 1% spenders with organic loyalty. |
| **Cluster 4** | Power Users | Extreme F / Low R; highly habituated (weekly) shoppers. |
| **Cluster 5** | Newbies | Low R / Low F; recent acquisitions with no history yet. |
| **Cluster 6** | Loyalists | High F / Mid M; consistent monthly revenue generators. |
| **Cluster 7** | Value High-Spenders | High M / High CR; wait for large promotions for bulk orders. |

*G. Predictive models (LightGBM and CatBoost)*

*1) Experimental environment setup*

To ensure an unbiased comparison, the implementation followed a standardized multi-layer architecture:

- Stratified experimental splitting: The dataset was partitioned into 80% Training and 20% Testing. Stratified Sampling was used in this research to maintain consistent churn rates across both subsets, preventing distribution bias.
- Balancing the scales (SMOTE): Through the use of SMOTE (Synthetic Minority Over-sampling Technique), the researchers were able to deal with an extreme class imbalance (the number of customers that "Stay" is substantially larger than those who "Churn"). Researchers used SMOTE to make "synthetic customers" based on what customers who have left look like. This taught the model the abstract boundaries of "at-risk" behaviour instead of just memorising the customers' history.
- Feature integration: The input for both models was a calculated mix of (i) Core RFM-C metrics, (ii) Behavioral history (Tenure, Complaints), and (iii) Latent structures (labels derived from the DP-Means algorithm).

*2) Parallel model execution and optimization*

We employed two separate training pipelines to take advantage of each tree algorithm's unique strengths:

- The LightGBM pipeline was created using a leaf-wise expansion method for improved training speed and variance reduction. The tree is constructed by adding leaves to any branch that was last used to split on the maximum loss, meaning you can "dig deeper" into each branch and, therefore, discover the most complex patterns. Categorical variables were also manually encoded into numerical data before training with this model.
- The CatBoost pipeline was created using symmetric trees (i.e., all branches grow equally) with a focus on providing stability as well as directly handling categorical features. Both Cluster_ID and coupon status can be processed natively within the framework of CatBoost, without using traditional encoding methods, preserving the original dataset's structure.

Optuna was utilized for the processes of Bayesian Optimization, rather than manual tuning or exhaustive grid searching. As a result of this, Optuna automatically found the "golden" parameters of both models to ensure that their best performance was achieved prior to the final comparison.

*3) Evaluation framework*

The more effective model is identified by not only simple accuracy, but also by three rigorous dimensions:

- Precision-recall curve (PRC): In churn prediction, there are far more customers who continue purchasing than those who leave. PRC helps to measure the model's ability to catch churners while maintaining high confidence in its prediction. A high-performing model must catch as many at-risk customers as possible (high recall) without causing false alarms (high precision).
- Computational complexity analysis: Even if a model is highly accurate can be useless in an e-commerce environment if it takes 10 minutes to produce a single prediction. Our group measured how long it takes the model to learn the entire dataset and the speed at which the model returns a result when a new customer transaction occurs. This is critical for real-world business use, where a company might want to send a retention offer the instant a customer shows signs of leaving.
- Evaluating XAI (Explainable AI) with a benchmarking technique: We utilized an interpretability method called SHAP (SHapley Additive exPlanations) to evaluate how two separate model architectures derive meanings from the same raw data. This is done to establish whether both algorithms recognize the same fundamental risk drivers (e.g. Complain or Tenure) and whether there is a divergence in the interpretation of the customer journey due to differences in their respective architectures.

# IV. results and discussion

## A. Descriptive statistics and RFM analysis

### 1) Purchase behavior across product categories

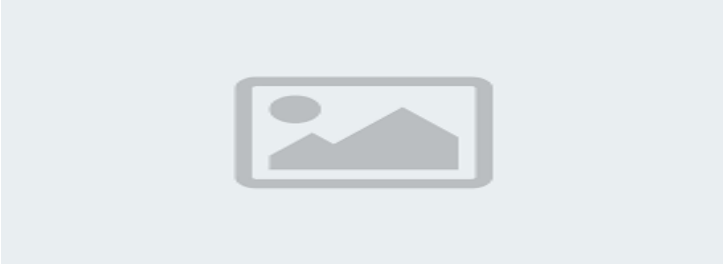

Fig.1. Sales concentration over product categories

Most orders are concentrated in a small number of product categories, with clear differences in customer activity across these categories.Customers buying popular products show higher activity, whereas those linked to less common products show lower activity.

### 2) Revenue structure and customer segmentation

Fig.2. Revenue distribution

Most customers spend small amounts, this is the regular type. In contrast, a small group of customers spends much more and contributes a large share of the total revenue which is VIP type.

Churn from this VIP group would have a much bigger impact on the business, while churn from regular customers is less critical. Because of this, it is more useful to focus on value-weighted churn rather than treating all customers the same. To support this, we add features Avg_Order_Value and Spend_last_30d in the data preparation for machine learning. For example, customers with high past spending but no recent activity are more likely to leave.

### 3) Churn pattern observation

This churn is taken from a separate dataset and is not connected to our main dataset, so it is used only for pattern observation.

Churn affects only a small share of customers, making patterns harder to observe. Tenure shows a clearer pattern. Early churn is more common in the first few months, whereas longer-tenure customers tend to remain active. This means that people are less likely to stick around at the beginning, when some customers leave before they become regular users.

These patterns are useful for understanding customer behavior, although they are explored separately since this dataset is not directly connected to the main transaction data and cannot be reliably combined.

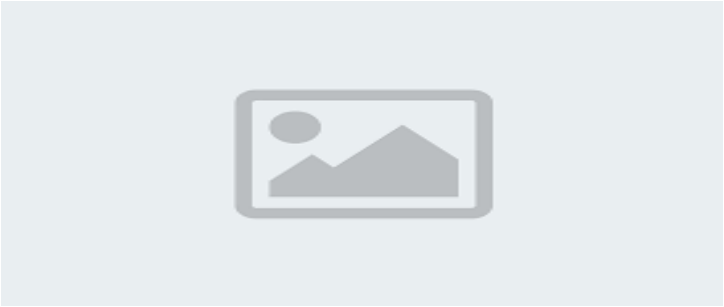

Fig.3. Customer churn rate

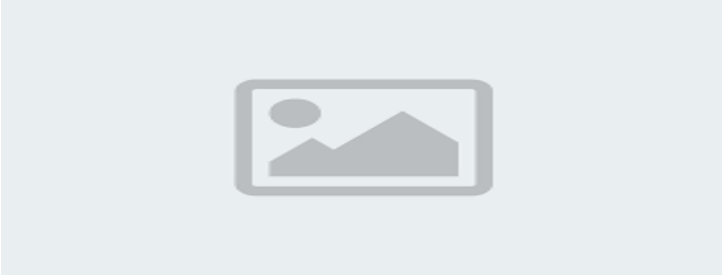

Fig.4. Customer Tenure vs churn

## B. Inferential statistics

### 1) Independence t-test

T-test findings indicate that the p-value is 0.00126. This value is less than the significance level of 0.05 and this implies that the null hypothesis is not accepted. Based on the visualization (bar chart of AOV) it is evident that there is a higher value in order amongst customers who use the coupons as opposed to those who do not.

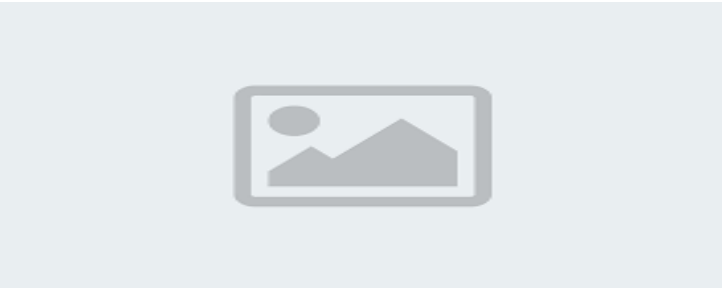

Fig.5. AOV by Coupon Usage

### 2) One-way ANOVA

In terms of the ANOVA test, the p-value 0.007 is statistically significant ($p<0.05$), which means that there are indeed significant differences in revenue across the different discount levels. This is confirmed by the view, which demonstrates different discount levels (10%, 20%, and 30%) have varying impacts on revenue, rather than a consistent impact.

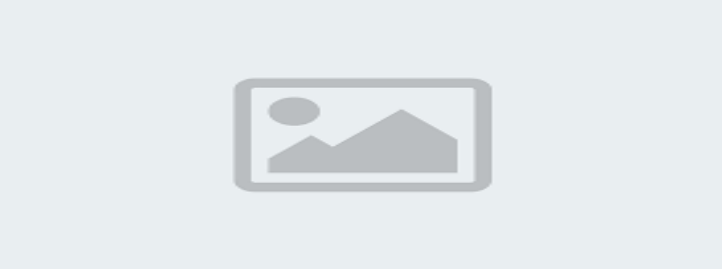

Fig.6. Revenue by Discount %

### 3) Chi-square test

In the Chi-square test, the p-value is very insignificant (2.66 1078). The stacked bar chart shows that the churn rate of customers who complain is by far on the higher side (31.7) than with the customers who do not complain (10.9).

Fig.7. Correlation analysis: Complain vs Churn

### 4) Discussion and insights

Based on these findings, it is apparent that promotions are highly influential in customer behavioral change in terms of expenditure. Coupons have the opportunity to boost AOV, a good move when revenue needs to be boosted in the short-term. A result of the ANOVA would indicate, however, that all levels of discounting are not equally successful and therefore a business must select the most appropriate level. More to the point, Chi-square outcome reveals that experience in the eyes of customers is a critical factor. Although promotions may boost expenditure, they will not avert churn in case customers are not satisfied. The complainations are good indications of dissatisfaction among customers and thus should be addressed promptly.

## C. Lasso Regression

### 1) Statistical results and feature importance

The Lasso model achieved an impressive AUC-ROC score of 0.98.

TABLE II. LASSO REGRESSION COEFFICIENTS ANALYSIS

| Feature | Lasso Coefficient | Analytical interpretation |
|---|---|---|
| **Frequency** | +0.2227 | Repeat purchase behavior is the absolute strongest indicator of a High-Value Customer. |
| **Recency** | -0.0555 | The longer the time since the last purchase, the lower the probability of being a High-Value customer. |
| **Monetary** | 0.0000 | Lasso determined this as redundant because its variance was already captured by Frequency, preventing multicollinearity. |
| **Coupon_Ratio** | 0.0000 | In this baseline, coupon dependency was treated as secondary noise compared to pure transaction frequency. |

### 2) Discussion and insights

The shrinkage of the coefficients to zero for Recency, Monetary, and Coupon Ratio is what makes the difference. The only positive predictive factor is the Frequency of coupon use (+0.2227). So, spending more or using more coupons does not predict future loyalty and is not as important as the actual return behavior.

This is against the grain of the traditional business thinking of "High Spenders only". The Lasso model suggests that the "habit of purchase" is a more useful definition of loyalty than an individual purchase.

## D. DP-Means

### 1) Experimental result and quality

DP-Means clustering was implemented to find natural customer segments which allows automatic cluster determination based on a penalty factor (λ). Silhouette and Elbow methods found the optimal value of λ was between 15-20 extracting 8 clusters.

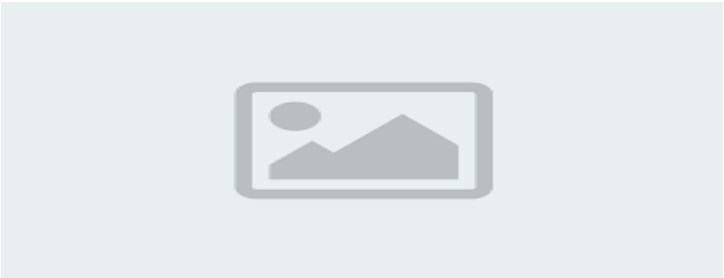

Fig.8. Elbow: Lambda vs Number of clusters

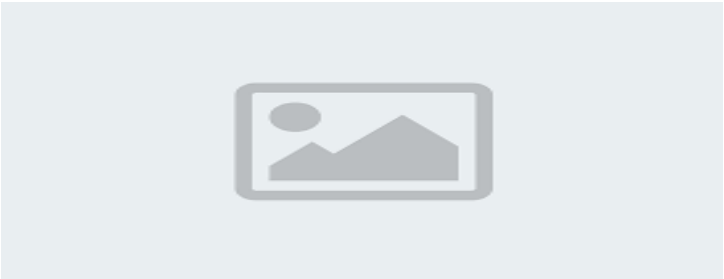

Fig.9. Elbow: Lambda vs WCSS

Fig.10. Silhouette plot

### 2) Discussion and insights

The algorithm is able to separate "Organic VIPs" from "Deal Hunters". The Silhouette Score (0.25) suggests good clustering. Clusters 1 (High Coupon Ratio) is a volume generator in the discounted period while Clusters 3 and 4 (Low Coupon Ratio) are organic high revenue generators.

- "Organic VIPs" provide an opportunity to reduce costs. There is no point in giving a customer a huge discount to buy.
- With "At-Risk" customers, coupons can be used only as a re-activation tool. The "Impact of Promotion" is not in strengthening true believers, but in re-awakening the sleeping hibernates.
- Sales teams will need to understand and talk to their customers: "Deal Hunters" want price promotions while "Champions" want rewards for being loyal: early access, or special services.

## E. Predictive models (LightGBM and CatBoost)

### 1) Performance evaluation

LightGBM does very well on the test set, with scores that are almost the best they can be.

- ROC-AUC is 0.9976, which shows that the model can separate churn and non-churn customers very well.
- PR-AUC is 0.9922, which shows that the model handles imbalanced data effectively. In churn prediction, PR-AUC is more important than ROC-AUC because it focuses more on churn customers.
- Precision is 0.9838, Recall is 0.9579, and F1-score is 0.9707. This means the model is both accurate and well balanced in detecting churn while reducing wrong predictions.

CatBoost shows slightly better results than LightGBM.

- ROC-AUC reaches 0.9999
- PR-AUC reaches 0.9993
- Precision, Recall, and F1-score all reach 0.9895

PR Curve of CatBoost is almost above LightGBM across the entire recall range, which shows better and more stable churn detection performance.

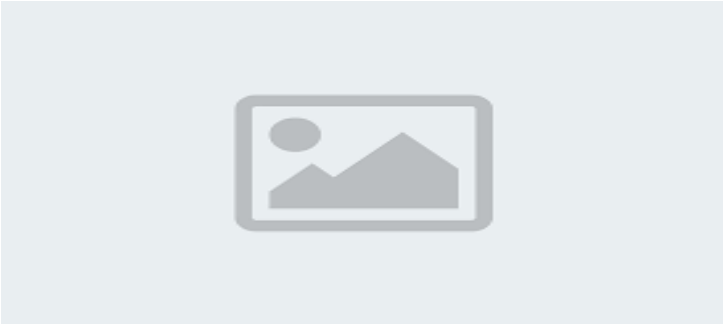

Fig.11. ROC curve

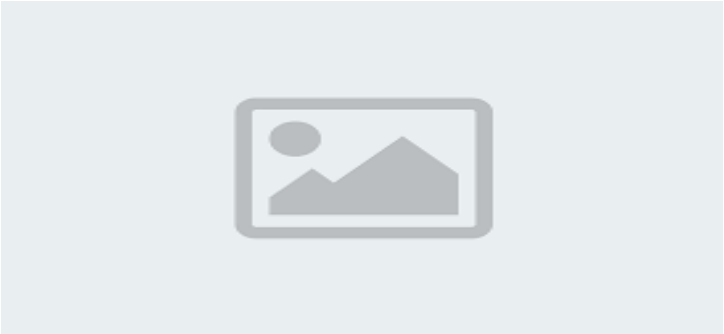

Fig.12. PR curve

## 2) Threshold optimization

LightGBM doesn't use the default threshold of 0.5. It uses the best threshold instead:

**Set the threshold to 0.4141**.

Lowering the threshold helps to :

- boost Recall
- lower the number of false negatives
- Don't forget about clients who leave

This result fits the shape of the PR Curve, where precision stays high for most recall values.

The optimal threshold of CatBoost is:

**Threshold = 0.1616**.

This threshold is much lower than LightGBM, which helps to:

- increase churn detection
- maintain high precision
- better support retention strategies

This shows that CatBoost has very strong prediction quality even with a low threshold

## 3) Interpretability analysis (SHAP)

### a) LightGBM

The most important features are:

- Tenure
- Complain
- NumberOfAddress
- CashbackAmount
- SatisfactionScore

Among them, Tenure is the most important factor.

- Low Tenure → higher churn
- Having complaints → higher churn

In addition, SatisfactionScore and DaySinceLastOrder also affect churn. Higher SatisfactionScore is linked to higher churn, although this is unusual. This may mean that short-term satisfaction does not always lead to long-term loyalty. Lower DaySinceLastOrder is also linked to higher churn. This shows that some customers may leave even after buying recently. Their buying behavior may be temporary, not stable.

**b) CatBoost**

The top features of CatBoost are:

- Tenure
- Complain
- NumberOfAddress
- SatisfactionScore
- CashbackAmount

The results show that:

- Low Tenure → higher churn
- Having complaints → higher churn
- Higher SatisfactionScore → higher churn
- Lower DaySinceLastOrder → higher churn

CatBoost shows a more stable SHAP distribution, especially for categorical features because it can handle categorical features directly without the need for encoding.

### 4) Model comparison

Both models achieve very high performance, but CatBoost performs slightly better in:

- PR-AUC
- Recall
- F1-score

LightGBM is better in training speed and tuning hyperparameters in a flexible way. CatBoost is better at dealing with categorical features and keeping models stable.Both models function well, however CatBoost is better at dealing with categorical data, and LightGBM converges a little faster. CatBoost is the better choice for forecasting turnover in this investigation.

### 5) Real business implications

Concentrate on the experience of new customers

- SHAP demonstrates that Tenure and Complain are the two things that have the biggest impact on churn.
- New clients don't normally stay with a company for long and are more likely to notice service flaws.
- If a consumer complains early on, they are likely to depart the service.

Know what makes you happy in the short term

- For some categories of customers, a high SatisfactionScore and a low DaySinceLastOrder are connected to more turnover.
- This means that people might be satisfied after buying something, yet they could still depart later.
- Being happy for a short time doesn't always indicate you'll stay loyal for a long time.

Money benefits are important

- CashbackAmount is one of the most crucial parts of the CatBoost model.
- Cashback, discounts, and other financial advantages have a big effect on some customers.
- These economic benefits will determine whether they stay or leave.

A technique for keeping customers proactively

- The best threshold is low (0.1616), which suggests that Recall is more important to the model.
- This helps find the danger of churn early, even when the signal is faint.
- Stopping consumers from leaving early is[4] generally less expensive than getting new ones.[4]

# V. conclusion and recommendations

## A. Conclusion

Through a careful process of testing and looking at many sets of e-commerce data, we have reached some crucial conclusions about modern customer behaviors and the relationship between coupons and loyalty. Most importantly, we have learned that customer loyalty will not be defined solely by static attributes (gender, location, etc.) but rather through the behaviors that customers demonstrate, such as the length of time they have purchased from a particular brand or the manner in which they access discount coupons.

We were also able to determine from our data that there exists a "high-risk" time frame for customers during the first few months of their relationship with the retailer. This is the time when customer engagement is very fragile and where attrition rates are at the highest point; however, once the customer has successfully surpassed this critical point, the likelihood of them becoming a long-term regular customer is multiplied many times.

Additionally, when we looked at the revenue distribution from a financial structural perspective, identifying the "heavy-tailed" revenue distribution has significantly changed our approach to risk management. The loss of one VIP customer through our use of DP-means clustering causes much more damage to cash flow than would the loss of one "regular" user.

Finally, while LightGBM has exhibited superior processing speed and computational efficiency, CatBoost was found to be much more sensitive in processing complex categorical variables. This allowed it to capture high-risk churn scenarios with greater precision without sacrificing the original integrity of the data features.

## B. Recommendations

Our strategic roadmap aims to deliver three pillar-based roadmaps for maximising revenue sustainability and brand health by providing the best service to our core customers through 1) transitioning to value-based retention instead of mass promotions, 2) implementing a first 90 days engagement programme, and 3) implementing CatBoost and LightGBM dual-engine technology for retrospective and real-time retention interventions.

In order to improve brand health and long-term revenue sustainability, it is vital to stop mass promotions and start focusing on value-based retention by investing marketing dollars into the VIP groups and using non-monetary perks such as preferential support/early access instead of monetary rewards as your means of differentiation.

To resolve early attrition, the organisation should establish a 90-day engagement plan that uses the first 90 days to establish customer loyalty via improved onboarding and behavioural nudges.

Thirdly, from an operational technical perspective, we recommend CatBoost for high-accuracy strategic planning at set intervals and LightGBM for real-time strategic planning to deliver instant, effective retention offers during live user interactions.

We do not recommend pursuing aggressive, large-scale investments in new (undefined) markets such as London and New York City. Instead, it is wiser to begin with small-scale A/B testing to uncover the specific "growth formula" for that region. By tightly integrating artificial intelligence with decisions based on actual behavioral data, the business can ensure it not only retains its most valuable customers but also optimizes every dollar of its marketing expenditure.

## C. Limitations

Even though the proposed models have high statistical significance, there are several limitations that should be recognized.

The study uses secondary transactional datasets, which document behavioral “symptoms" (such as a period of 250 days devoid of activity), but not the causes. Without emotional data or direct feedback, models will be unable to discern a lack of satisfaction with a service from competitor lure.

The analysis overlooks external market conditions, such as macro inflation, competitor pricing, or seasonal trends, all of which impact purchase frequency and consumer loyalty.

Another boundary of this study is that we are using the past to navigate the future. However, in the fast-paced e-commerce[6] landscape, consumer habits change every day. A static model that is not fed with a real-time data stream can only provide accurate answers to questions that are no longer relevant. This leads to a requirement that these models need to be[5] retrained periodically to maintain their accuracy.[5]

*D. Future research*

The best chance is to close the gap between transactional and motivational. Adding customer feedback and satisfaction surveys could make the RFM-C model better. Knowing how customers feel about spending will assist explain why happy customers who use coupons and like the products leave. Customers may still leave even if they are very happy with the service because of service problems, money problems, or a deal-oriented lifestyle.

To surpass the static constraints of current models, forthcoming research must employ dynamic models that adjust to evolving contexts. Recurrent neural networks, such as LSTM or GRU networks, could help you figure out how many customers are leaving in real time instead of using static regressions. Companies can use this kind of data to send out coupons when customers stop being interested in them.

Besides binary classification (Churn versus Non-Churn), further research will consider the profitability of the clients. Concentrate on forecasting CLV modeling. With Gamma-Gamma or Beta-Geometric models (BG/NBD), you will know how valuable a customer is for the next two years. If you know how to precisely calculate the Targeted CAC (Customer Acquisition Cost), you can make sure that coupon investments never go over the expected return from a segment. We also need to figure out the right amount of discount that will get them to buy without losing money. This lets you use "scientific discounting," which means giving one group 5% and another 15% based on what they require.

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